

HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination — Set C (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

Duration: 120 minutes

Total credit: 120 points

Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: _____

Matriculation no.: _____

Problem	P1	P2	P3	P4	P5	P6	P7	Total
Maximum points	16	8	20	16	20	20	20	120
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e , logarithms, $(0.95)^{40}$, t -quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines

(16 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) [4 P] A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at each knot. How many degrees of freedom (free parameters, counting the intercept) does a cubic spline with K interior knots use? Justify your answer by counting the parameters of the polynomial pieces and subtracting the constraints at the knots, and evaluate for $K = 3$.
- (b) [4 P] How many degrees of freedom does a *natural* cubic spline with the same $K = 3$ knots use? State the additional constraint a natural spline imposes, say in which region of the predictor space it takes effect, and explain why it is useful in practice.
- (c) [4 P] A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt.$$

Explain the role of each of the two terms, and describe the behaviour of the solution $\hat{g}$ in the two limiting cases $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.

- (d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ , define the truncated-power basis function explicitly, and explain in one sentence why adding this basis function does not destroy the smoothness of the fit at ξ .

Problem 2: Generalised additive models

(8 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \dots, X_p$, and explain in one sentence how it generalises multiple linear regression.
- (b) **[3 P]** State one important *advantage* of GAMs concerning the interpretation of the fitted functions f_j , and the main structural *limitation* of the additive form — including how this limitation can be repaired.
- (c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Problem 3: Regression and classification trees by hand

(20 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

A logistics company records for $n = 6$ regional depots the number of delivery vans in operation x and the number of parcels delivered that day y (in 1,000 parcels):

Depot i	1	2	3	4	5	6
Vans x_i	1	2	3	4	5	6
Parcels y_i	3	5	4	9	11	10

A regression tree with a *single* split at cutpoint s partitions the data into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

- (a) **[8 P]** For each of the two candidate cutpoints $s = 2.5$ and $s = 3.5$, determine the two regions, the region means and the resulting total residual sum of squares $\text{RSS}(s) = \sum_{m=1}^2 \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting choose, and why?
- (b) **[2 P]** Using the chosen split at $s = 3.5$ from (a), predict the number of parcels for a new depot that will operate $x = 5$ vans.
- (c) **[6 P]** A *classification* tree for a different task has a terminal node containing 9 training observations from three product categories: 5 “Electronics”, 3 “Clothing” and 1 “Books”. Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node (use $\ln(5/9) = -0.5878$, $\ln(1/3) = -1.0986$ and $\ln(1/9) = -2.1972$). Which class does the node predict?
- (d) **[4 P]** In *cost-complexity pruning* one minimises, for a subtree $T \subseteq T_0$ of the large tree T_0 ,

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|.$$

Explain the role of the tuning parameter α : what happens for $\alpha = 0$, what happens as α increases, and how is α chosen in practice?

Problem 4: Ensembles — bagging, random forests, boosting**(16 points)**

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

- (a) **[6 P]** Let $\hat{f}_1(x), \dots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x , each with variance σ^2 and with pairwise correlation ρ . The variance of the bagged average is

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)\right) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2.$$

- (i) Explain briefly why this formula shows that bagging reduces variance, and which term limits the reduction. (ii) Evaluate the variance for $B = 50$, $\rho = 0.4$ and $\sigma^2 = 16$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \rightarrow \infty$?
- (b) **[4 P]** How do *random forests* modify bagging at each split, and what is the recommended subset size m for *regression* as a function of p (evaluate for $p = 30$)? Using the bagged-variance formula $\text{Var} = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2$ from (a), explain why this modification improves on bagging.
- (c) **[4 P]** A colleague fits a boosted regression-tree model in scikit-learn with the call below; its three arguments are exactly the three tuning parameters of *boosting*:

```
from sklearn.ensemble import GradientBoostingRegressor
boost = GradientBoostingRegressor(
    n_estimators=5000, learning_rate=0.01, max_depth=1)
boost.fit(X_train, y_train)
```

For each of `n_estimators`, `learning_rate` and `max_depth`, state in one sentence what it controls, and explain the “slow learning” idea behind boosting: why is a small `learning_rate` paired with a large `n_estimators`?

- (d) **[2 P]** What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it available essentially for free?

Problem 5: Neural networks by hand**(20 points)**

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Consider a tiny feed-forward network for a quantitative response with input $x = (x_1, x_2) = (2, 3)$, one hidden layer with two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g(-3 + 2x_1 + 1x_2), \quad A_2 = g(2 - 2x_1 - 1x_2), \quad f(x) = -2 + 3A_1 + 2A_2.$$

- (a) **[8 P]** Perform the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the network output $f(x)$. Comment briefly on what the ReLU did to the second unit and why such non-linear activations are essential.
- (b) **[6 P]** A classification network with $K = 3$ classes produces the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (2.0, 0.0, -1.0)$ for one observation whose true class is class 2. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. (Use $e^2 = 7.389$, $e^0 = 1.000$, $e^{-1} = 0.368$ and $\ln 8.757 = 2.170$.)
- (c) **[4 P]** Count the parameters of a fully connected network with $p = 6$ inputs, one hidden layer with $k = 5$ units and an output layer with $K = 3$ units, where every hidden and output unit has a bias. Show the count separately for the two weight layers.

- (d) **[2 P]** A convolutional layer receives a 64×64 image and uses filters of size $F = 3$ with padding $P = 1$ and stride $S = 2$. Compute the spatial size of the output feature map from $\lfloor (W - F + 2P)/S \rfloor + 1$ (note the stride is 2, so the division is rounded down).

Problem 6: Multiple testing

(20 points)

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

- (a) **[4 P]** A lab performs $m = 40$ independent hypothesis tests, each at level $\alpha = 0.05$, and suppose all 40 null hypotheses are true. Define the family-wise error rate (FWER) and compute it for this situation (use $(0.95)^{40} = 0.1285$). Interpret the result in one sentence.
- (b) **[7 P]** A research group screens $m = 6$ candidate biomarkers; the ordered p -values are

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state for each method exactly which hypotheses are rejected, and explain why Holm can never reject fewer hypotheses than Bonferroni.

- (c) **[6 P]** Apply the *Benjamini–Hochberg* (BH) procedure with FDR level $q = 0.05$ to the same six ordered p -values (0.001, 0.008, 0.011, 0.030, 0.041, 0.260; $m = 6$): give the thresholds jq/m , determine which hypotheses are rejected, and state precisely what “FDR control at $q = 0.05$ ” guarantees for the resulting set of rejections.
- (d) **[3 P]** A pharmaceutical company runs a *confirmatory* phase-III trial that tests $m = 5$ pre-specified endpoints; a single false-positive endpoint could lead to the approval of an ineffective drug. Should they control the FWER or the FDR? Justify your recommendation.

Problem 7: Cumulative essentials (Chapters 2–6)

(20 points)

Lecture slides: Chapters 2–6 (Lectures 1–8).

- (a) **[4 P]** The *bootstrap* draws a resample of size n with replacement from the n observations. (i) Show that the probability that a given observation is *not* selected into a particular bootstrap sample equals $(1 - 1/n)^n$, and give its limit as $n \rightarrow \infty$ (use $e^{-1} = 0.368$). (ii) Hence what fraction of the original observations appears (at least once) in a typical bootstrap sample as $n \rightarrow \infty$? (iii) Compute this inclusion probability exactly for $n = 5$ (use $0.8^5 = 0.328$).
- (b) **[4 P]** At a fixed test point x_0 the expected test error of an estimate $\hat{f}$ decomposes as $E[(y_0 - \hat{f}(x_0))^2] = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\varepsilon)$. A simulation stores these three components for two candidate estimators in a `pandas DataFrame` and prints it:

```
comp = pd.DataFrame({"var": [3.5, 0.5], "bias": [-1.0, -2.5],
                    "noise_var": [2.0, 2.0]},
                    index=["flexible_spline", "linear_model"])
print(comp)
```

	var	bias	noise_var
flexible_spline	3.5	−1.0	2.0
linear_model	0.5	−2.5	2.0

Reading the three components off this output, compute the expected test MSE of each estimator, say which you would deploy at x_0 , and state the lowest test MSE *any* method could achieve here.

- (c) **[3 P]** Write down the *elastic net* penalty and explain what it inherits from the lasso and what it inherits from ridge regression. In which situation does the elastic net clearly outperform the lasso?
- (d) **[3 P]** Two predictors in a regression have correlation 0.95. Explain what such *collinearity* does to the least-squares coefficient estimates. The variance inflation factor is $VIF_j = 1/(1 - R_j^2)$, where R_j^2 is from regressing X_j on the other predictors; compute and interpret VIF_j for $R_j^2 = 0.9$ (use $\sqrt{10} = 3.162$).
- (e) **[6 P]** Match each scenario to exactly one method from $\{\textit{linear regression}, \textit{lasso}, \textit{logistic regression}, \textit{random forest}\}$ (each method used exactly once) and justify each choice in one sentence:
- (i) Estimate, with a 95% confidence interval, how much one additional year of schooling raises hourly wage, from $n = 3,000$ workers and a handful of predictors.
 - (ii) From $n = 80$ patients with $p = 8,000$ metabolite measurements, predict a continuous biomarker level and identify a short list of relevant metabolites.
 - (iii) Predict whether a customer will churn (yes/no) and report to management how each customer characteristic changes the *odds* of churning.
 - (iv) Forecast next-day electricity demand (continuous) from hundreds of weather and calendar variables with strong non-linear effects and interactions; only forecast accuracy matters.

End of the examination — good luck!